

From Riemann Hypothesis Formalization to the Empirical Foundations of Pat: Transformer OOD Observations, Structural Laws, and Machine-Checked Continuum Theorems

Anonymous Author(s)

Abstract

We record the route from the formalization of the Riemann direction to the empirical foundations of the Pat framework. The route begins with a Riemann-direction visualization and its Lean formalization (claims C011–C025, zero sorry), whose basepoint-drift intuitions motivate controlled transformer training observations; structural laws are extracted, and the route ends in machine-checked theorems. The starting observation is a two-pole out-of-distribution (OOD) dichotomy, reproduced in 45 training runs: a low-epoch transformer reaches training accuracy 1.000 on successor-notation arithmetic, yet OOD accuracy is 32/32 on length extrapolation in the trained notation and 0/32 on the same structure written in an untrained notation of the same basepoint; adding explicit equivalence statements bridges the two notations ($0/32 \rightarrow 32/32$). From this and supporting experiments we extract five structural laws (anchor binding, notation translatability, shift invariance, structure-bearing representation, definition-layer invisibility). The laws map, as an observational correspondence, to the Pat framework’s paired-direction declaration, phase-relation locking, and phase-magnitude interlock. A completeness criterion — the decoupling operator D (exclusion + cancellation + layering) implying completeness at any order, element, and subject-object — organizes the correspondence, and three mapping judgments (symbol-norm, presentation-legality, intuition-precision) give the layer-to-layer verdicts that the OOD two-pole dichotomy exhibits. The framework chain is formalized in Lean 4 / mathlib: twenty-two theorems with zero errors and zero sorry, from the interlock matrix (determinant $\theta \cdot (r+1/r) \neq 0$) through the four-phase interlock, the countable pat grid, and the density theorem, to the continuum closure theorems (the continuum $[0, 2\pi]$ is the closure of the pat grid; an endogenous version constructs the same closure from two paired basepoints and midpoint reduction). The honest boundary is stated exactly: the empirical-to-framework mapping is an observational correspondence, not a deductive implication, and the training results are reproducible from the artifact (one command, Python + PyTorch).

Keywords: Lean, formalization, transformer, out-of-distribution, pat grid, continuum, empirical

Verification map

Main theorems	Continuum closure: every $x \in [0, 2\pi]$ lies in the endogenous constructible grid; interlock: the phases: resolvability verdicts.
Assumptions	Lean 4.32.2 + mathlib v4.32.2; standard Lean axioms, the mathlib dependency; no additional axioms.
Core lemma chain	R143 interlock ($\det \neq 0$) $\rightarrow$ R149 four-phase interlock $\rightarrow$ R163 endogenous continuum closure
Lean theorems	MutualLocking.lean (3), Pat4Phase.lean (4) (5), ContinuumPatGrid.lean (1), MappingJudgmentTheorems.lean (5) — 22 total
Repository / version	artifact in this submission; lake build passes; minimal dependency closure.

1 Introduction

The Pat framework is a family of claims about direction declaration, phase locking, and symmetric-pair reduction, developed in a program of small transformer experiments. Small transformers trained on algorithmic datasets exhibit rich and counterintuitive generalization behavior: delayed generalization (grokking) on small algorithmic datasets [13], systematic failures on out-of-distribution inputs [5, 10], and length extrapolation that depends on the position encoding [6, 14, 16]; in-context arithmetic emerges at scale [2]. This paper records the empirical route that grounds the framework: what was observed, what structural laws were extracted, and which claims were machine-checked. The route is a chain of four links.

Route origin: the Riemann direction. The route begins in the Riemann direction: a zeta visualization project (2026-08-06, 138,067 zeros with the Riemann–Siegel correction, critical line and prime positions) and its Lean 4 / mathlib formalization (claims C011–C025, completed 2026-08-12, zero sorry). The formal work produced the basepoint-drift intuitions that motivate the empirical part: primes have no

construction, irrationals are projected remainders, and basepoints drift on the circle. These intuitions are tested in controlled transformer experiments (Section 2), which extract the structural laws (Section 3). The Riemann hypothesis itself is not claimed here; the honest boundary is stated in Section 8.

Empirical starting point. A low-epoch transformer trained on successor-notation arithmetic (addition up to 9, multiplication up to 4, fully enumerated with true and false labels) reaches training accuracy 1.000 in every one of 45 runs (5 groups $\times$ 3 seeds $\times$ 3 independent executions). Yet out-of-distribution (OOD) accuracy dichotomizes: 32/32 on length extrapolation in the trained notation, 0/32 on the same structure written in an untrained second notation of the same basepoint, and 32/32 after explicit equivalence statements bridge the two notations. Training completeness does not imply extrapolation completeness; this is the starting observation from which everything else follows.

Structural laws. From this and the supporting experiments we extract five laws about what training intuition is and is not: operations bind to a basepoint (anchor binding); notations of the same anchor are translatable when equivalence is declared (notation translatability); one step of successor extrapolates along the chain (shift invariance); structure must be fully expanded in the sequence (structure-bearing representation); and the definition layer of tokens is invisible to training (definition-layer invisibility). The meta-law is: training intuition is representation-bound statistics.

Framework correspondence. The laws correspond, as an observational correspondence rather than a deductive implication, to framework claims: anchor binding to paired basepoint declaration (R136), notation translatability to symmetric-pair reduction (R143), and the collapse of undeclared notations to the self-referential degenerate case (R138). A completeness criterion organizes the correspondence: the decoupling operator D — exclusion, cancellation, layering — implies completeness at any order, any element, and any subject-object, exhibited as perfect OOD behavior; and three mapping judgments (symbol-norm, presentation-legality, intuition-precision) give the layer-to-layer verdicts that the OOD two-pole dichotomy exhibits.

Machine-checked theorems. The framework chain is formalized in Lean 4 / mathlib: twenty-two theorems, zero errors, zero sorry, from the interlock matrix to the continuum closure theorems. The theorem chain is: the interlock matrix is nonsingular ($\det = \theta \cdot (r + 1/r) \neq 0$); the four phases $\{1, -1, i, -i\}$ pairwise interlock; every phase in $[0, 2\pi]$ is uniformly approximated by the countable pat grid $\{2\pi j/N\}$; the continuum is the closure of the pat grid; and an endogenous construction from two paired basepoints and midpoint reduction reaches the same closure.

The paper is organized as the route itself: Section 2 reports the empirical starting point; Section 3 states the structural laws; Section 4 gives the framework correspondence; Section 5 states the completeness criterion and the mapping judgments; Section 6 presents the machine-checked theorems; Section 7 describes the exercise series; Section 8 states the honest boundary; Section 9 concludes.

2 The empirical starting point

2.1 Setup and judgment discipline

All experiments use the same training runtime (included in the artifact), a small transformer [17] (dimension 64, 2 layers, 60 epochs, learning rate 10^{-3} , batch 512) over tokenized arithmetic sequences. Tokens are fully expanded successor chains (true numerals): the chain length carries the number. The judgment discipline is strict: a sample is correct only if the full sequence is reconstructed token by token (end-truth judgment); positional accuracy is not trusted, because aggregate accuracy masks collapse points. Every reported count is a full-sequence count.

2.2 Observation 1: training accuracy 1.000, OOD dichotomizes

Training completes perfectly in all 45 runs (5 groups $\times$ 3 seeds $\times$ 3 executions) — no exception. OOD, however, splits into two poles: some probes score 32/32, others 0/32. Training completeness $\neq$ extrapolation completeness. This dichotomy is the fixed point from which all subsequent observations are made.

2.3 Observation 2: same-notation extrapolation is 32/32

Group 1a: train on the notation $succ_a$ (addition ≤ 9 , multiplication ≤ 4), probe OOD by length extrapolation in $succ_a$ (addition 10–12, multiplication 5–6): 32/32. The chain-length rule (chain concatenation) holds beyond the training coverage. Within a representation, statistics extrapolate; this is consistent with prior reports of length generalization in arithmetic transformers [6, 10].

2.4 Observation 3: same-basepoint zero-shot is 0/32

Group 1b: the training samples are identical to 1a; only the OOD probe changes notation, from $succ_a$ to $succ_b$ — the same successor structure, same basepoint, a different spelling. The probe scores 0/32: complete collapse. $succ_b$ is “the same successor written differently”; zero-shot does not transfer. Training intuition binds to the notation, not to the semantics: the meaning of *successor* never entered the training.

2.5 Observation 4: equivalence statements bridge, 0/32 to 32/32

Group 2: add equivalence contrast samples to the training set ($succ_a$ -chain $n = succ_b$ -chain n , $n = 0..9$) plus anchored

wrong samples (shown by a control group to be irrelevant).
The OOD probe on $succ_b$ now scores 32/32. Explicit state-
ments that the two notations are equivalent bridge the no-
tations: $0/32 \rightarrow 32/32$. An equivalence statement is a paired
declaration; it is the bridge tool.

2.6 Reproducibility

All three executions (two in the original environment, one
inside the self-contained package) reproduce the same per-
seed counts: 32/32, 0/32, 32/32 for groups 1a, 1b, 2 respec-
tively, with the control groups 2' (bridge does not damage
the original notation) and 4 (symmetry-special training) also
32/32. The core two-pole contrast ($0/32 \leftrightarrow 32/32$) is stable
across seeds and executions.

3 Five structural laws

3.1 Law 1: anchor binding (E11)

Shifting the basepoint (anchor drift) collapses zero-shot to
0/32; each anchor requires its own training; explicit equiv-
alence samples between anchors do not transfer the oper-
ation rules. Anchor type makes no difference (a value-zero
anchor and a basepoint anchor drift identically). Every refer-
ence frame's operations must be established independently.

3.2 Law 2: notation translatability (E12)

Different notations of the same anchor are bridgeable by
equivalence statements ($0/32 \rightarrow 32/32$); the same notation
across anchors is not bridgeable (E11, 0/32). The anchor (base-
point) is the structural identity to which operations bind; a
notation is a translatable representation within an anchor.
Contrast samples (equivalence statements) are the bridge
tool.

3.3 Law 3: shift invariance (E9)

Training only $succ(0) = 1$ (two samples) extrapolates to
 $succ(1..7)$: 14/14. The step $0 \rightarrow 1$ and the step $1 \rightarrow 2$ are
the same successor; this is the mechanism of chain-length
extrapolation.

3.4 Law 4: structure-bearing representation (E6)

Position-value bare symbols (implicit place weight, nomi-
nal position tokens) collapse (0/30); true numerals (fully ex-
panded successor chains, chain length = value) are stable
(30/30). Structure must be fully expanded in the sequence;
it cannot be left to intuition.

3.5 Law 5: definition-layer invisibility (E4/E5)

Four definitions of 0 (postulate, iteration zero, empty-chain
default, cancellation law) and two designs of successor (ex-
plicit concept, atomic counting) train equivalently. Training
intuition is sensitive only to the representation form (se-
quence structure); the token definition layer (where the se-
mantics comes from) is invisible.

3.6 The meta-law: training intuition is representation-bound statistics

Strip one representation and probe the same value in an-
other: total collapse (0/32). Joint training on several repre-
sentations does not induce cross-representation invariants
(a third representation still scores 0/32). An OOD score of
1.0 proves only that statistics extrapolate within that repre-
sentation; it says nothing about structure. Structure (value,
semantics) is the object of the formalization layer (Lean, def-
initions); the training layer cannot reach it.

4 The framework correspondence

The five laws correspond to framework claims. We state the
correspondence explicitly and mark it as observational: each
framework concept has a statistical manifestation, but the
manifestation does not imply the concept.

Anchor binding $\leftrightarrow$ *paired basepoint declaration* (R136).

Empirically, operations bind to an anchor: shifting it col-
lapses (0/32), changing notation without declaration collapses
(0/32). In the framework, a basepoint cannot be used alone;
directions must be declared in symmetric pairs at once (a sin-
gle direction is a privilege pollution). The empirical 0/32 is
the statistical manifestation of an undeclared pair: training
 $succ_a$ without declaring $succ_b$ leaves the $succ_b$ side without
an anchor, and it collapses.

Equivalence statements $\leftrightarrow$ *symmetric-pair reduction*
(R143). Empirically, equivalence contrast samples bridge two
notations ($0/32 \rightarrow 32/32$). In the framework, a symmetric
pair $\{succ_a, succ_b\}$ reduces to 1: the numerical pair $r \cdot (1/r) =$
1 and the phase pair $e^{i\theta} \cdot e^{-i\theta} = 1$ are the two decomposi-
tions of the unit. The bridge is the statistical manifestation
of symmetric-pair reduction: once declared, $succ_b$ is solvable
from $succ_a$ and back.

Interlock: bidirectional solvability (R143). After bridg-
ing, both directions work (groups 2 and 2', both 32/32). In
the framework, interlock is equivalent to nonsingularity of
the interlock matrix ($\det = \theta \cdot (r + 1/r) \neq 0$): from either
coordinate (phase or magnitude) the other is solvable. Bidi-
rectional translatability is the statistical manifestation of in-
terlock nonsingularity.

Undeclared = collapse (R138). An undeclared equiva-
lence collapses to 0/32: information is unreachable across
representations. In the framework, an unlocked phase re-
lation is the self-referential degenerate case: every opera-
tion on it returns itself (absorption). The 0/32 is the statisti-
cal manifestation of the degenerate collapse: the undeclared
side's information is unreachable, and the loop closes on it-
self.

Shift invariance $\leftrightarrow$ *successor identity* (R137). Empiri-
cally $succ(0)$ extrapolates to $succ(1..7)$. In the framework, a

declared direction yields a non-collapsing chain: each successor step is the same direction. Chain-length extrapolation is the statistical manifestation of chain injectivity.

5 The completeness criterion and the mapping judgments

5.1 The completeness criterion

The correspondence above is organized by a completeness criterion: the decoupling operator D — exclusion (remove what does not belong), cancellation (pair off and cancel what reduces), layering (separate what plays different roles) — implies completeness at any order, any element, and any subject-object, exhibited as perfect OOD behavior. The empirical program verifies the criterion's clauses: at the order level, algebraic axioms 12/12, involutions 12/12, truth/false separation 6/6 (token-level, per the reporting discipline); at the element level, the orthogonality-declaration bridging succeeds same-anchor (32/32) and a double declaration (orthogonality + equivalence) restores stability across seeds (5/5); at the subject-object level, direction localization with two symmetry groups scores 16/16 $\times$ 3 seeds, and a single declaration is insufficient — two groups of symmetries are necessary to localize a direction (Cartan–Dieudonné: rotations are composites of two reflections; a single reflection only defines $\pm$).

5.2 The three mapping judgments

The OOD two-pole dichotomy is the training-layer exhibition of three layer-to-layer judgments about the token system (five layers: basepostulate B, concept C, symbol S, grammar G, presentation P). Each judgment decides, for a multi-to-multi mapping between layers, whether every target is resolvable.

Symbol-norm judgment ($S \rightarrow C/I$). How is the multi-to-multi mapping from symbols to logic and intuition judged? Every target must be resolvable: ambiguity converges by context (no context = all candidates; given context = one); unresolvable ambiguity or collision is non-norm. The two symmetric directions are symbol $\rightarrow$ logic and symbol $\rightarrow$ intuition. The experimental exhibition is the structure-vs-symbol contrast: mapped symbols 30/30, bare symbols 0/30 (E6).

Presentation-legality judgment ($C/I \rightarrow P$). How is the multi-to-multi mapping from logic and intuition to presentation judged? Every presentation source must be resolvable: multiple presentations of one source are bridgeable by equivalence statements; numeric coincidence without explicit declaration is illegal (the supervision signal carries no direction information). The experimental exhibitions are the two poles of the dichotomy: undeclared notation collapse (0/32) and equivalence bridging (32/32).

Intuition-precision judgment ($I \rightarrow C/P$). How is the multi-to-multi mapping from intuition to logic and presentation judged? The intuition-token interface must be definitionally resolvable (it must not impersonate a logic concept); intuition-path precision is judged by the two-pole comparison (in-representation OOD 1.0, cross-representation 0); polarity confusion events are the canonical failure. The experimental exhibition is exactly the two-pole dichotomy with the bridge as the resolvability restoration (0/32 $\rightarrow$ 32/32).

A fourth judgment (symbol $\rightarrow$ presentation: notation context precedence and associativity resolve; glyph identification is unique; round trips are lossless) completes the five-layer coverage. All four judgments share one structure: resolvability of every target of a multi-to-multi mapping, judged by declaration, context, or two-pole comparison.

6 Machine-checked theorems

The theorems are formalized in Lean 4 [3] against mathlib [4, 11]. Recent systems couple language models with proof assistants for formal mathematics — from curriculum learning over formal statements [7, 12] and the MiniF2F benchmark [20], to retrieval-augmented and copilot-style proving [15, 19], interleaved thinking and proving [9], large-scale synthetic data [1, 18], and guiding provers with informal proofs [8]. The present development is a small, fully machine-checked library produced in this style (Section 8).

6.1 The interlock matrix (R143, MutualLocking.lean)

A declaration is two groups of symmetries: a phase (direction) pair $(\theta, -\theta)$ and a magnitude (distance) pair $(r, 1/r)$. The two paired vectors are a 2×2 matrix

$$\begin{pmatrix} \theta & r \\ -\theta & 1/r \end{pmatrix}, \quad \det = \theta \cdot (r + 1/r).$$

`mutual_lock_invertible`: for $\theta \neq 0$ and $r > 0$ the determinant is nonzero — the matrix is nonsingular, so either coordinate is solvable from the other. Phase and magnitude are mutually locked; no third layer of locking is needed. Supporting theorems: `magnitude_locks_phase_round_trip` (the normalized displacement $(pat0+d-pat0)/\|pat0+d-pat0\| = d$ recovers the direction from the locked magnitude position) and `magnitude_pair_log_mirror` ($\log(1/r) = -\log r$: the magnitude pair is the log mirror pair).

6.2 Four-phase interlock (R149, Pat4Phase.lean)

The unit's two symmetry decompositions combine: the numerical pair $a \cdot (1/a) = 1$ and the phase pair $e^{i\theta} \cdot e^{-i\theta} = 1$. `quadrphase_interlock`: the four phases $\{1, -1, i, -i\}$ pairwise interlock (magnitude pair + phase pair), normalizing to S^3 . Supporting theorems establish axis orthogonality (`axis_pair_orthog`) and the extrapolation to the pat circle (`extrapolation_to_pat_circle`, `infinite_isomorphic_extrapolation`).

6.3 Countable grid and density (R150, PatCountableInfinPhaseUnification.lean)

The pat grid is the union of all N -slot rings:

$$\text{patGrid} = \{ 2\pi j/N \mid 0 < N, j \leq N \}.$$

pat_grid_countable: the grid is countable (it is the image of $\mathbb{N} \times \mathbb{N}$). pat_phase_unification: every phase $\theta \in [0, 2\pi]$ is approximated to arbitrary precision by a grid point ($\forall \epsilon > 0, \exists x \in \text{patGrid}, |\theta - x| \leq \epsilon$); the quantization error is $\leq \pi/N$, so $N \geq \pi/\epsilon$ suffices. This is the bridge theorem from the finite construction to the continuum.

6.4 Continuum closure (R151, ContinuumPatGrid.lean)

continuum_in_pat_grid_closure: every $x \in [0, 2\pi]$ belongs to the closure of the pat grid. The proof uses the density theorem and the closure criterion directly (no sequence or limit lemmas). The continuum is not a clean object; it is the closure of the countable reachable grid — a generated structure's limit.

6.5 Endogenous closure (R163, ContinuumConstructible.lean)

The closure is re-derived from endogenous generation: two paired basepoints ($\text{base}_0 = 0, \text{base}_1 = 2\pi$) and the symmetry reduction rule (the reduction point of a symmetric pair is its midpoint) generate the grid

$$\text{patGrid}' = \{ 2\pi \cdot k/2^m \mid 0 \leq k \leq 2^m \}$$

(dyadic rationals $\times 2\pi$). dyadic_InPat: every dyadic point is generated; patGrid'_dense: the generated grid is dense in $[0, 2\pi]$; continuum_in_constructible_closure: every $x \in [0, 2\pi]$ lies in the closure of the generated grid. The definition layer uses no natural numbers: generation is closure under midpoint from the two paired basepoints, and the dyadic representation is a representation theorem, not an input. The construction mirrors the paired-declaration discipline: two basepoints, declared as a pair; symmetric pairs reduce to their midpoint.

6.6 Mapping judgments (MappingJudgmentTheorems.lean)

Five theorems formalize the resolvability verdicts of Section 5: context_convergence_resolvable (context convergence implies the existence of a resolution verdict for the symbol mapping), equivalence_bridge_judgment (an equivalence declaration yields a bridge verdict), conflict_declared_resolvable (a declared conflict yields an injective resolvability verdict), round_trip_precise_injective (lossless round trip $B \circ A = \text{id}$ implies the assembly map is injective — symbols are resolvable), and polar_judgment_separated (the two-pole judgment separates: in-representation truth and cross-representation falsity are judged by the same verdict structure). All five compile with zero errors and zero sorry.

7 The exercise series

The route is accompanied by an independent verification face: the homework-exercise series — twenty-five observation reports (Exercises I–XXII, the Pat0 exercise, and the series overview), each with its own Lean formalization, all zero sorry. The method of the series is the “0pat” discipline: take a pat observation, walk the genealogy map from the pat concept to its standard- mathematics coordinate, formalize the equivalence along the way, and keep the pat vocabulary out of the proof. Each exercise is a small independent formalization; together they form a corpus of machine-checked statements that the framework's concepts reduce to. The full series (reports, TeX sources, Lean code, and overview) is included in the artifact under exercises/. The series also records the oldest empirical anchor of the route: the first generalization observation, dated 2026-07-29 in the session log.

8 Honest boundary

The boundary is stated exactly, in four points.

The empirical-to-framework mapping is observational.

The correspondences of Section 4 (statistical manifestations) are not deductive implications: no theorem states that training behavior implies a framework claim. The Lean theorems hold independently of the training observations; the observations provide motivation and correspondence.

The Riemann hypothesis is not claimed. The Riemann direction is the route's origin and motivation; its formalization (C011–C025) covers the basepoint-drift structure of the direction. No theorem here claims the Riemann hypothesis itself; it remains the unproved endpoint of that direction, out of scope for this paper.

No structural claim about the model. Training intuition is representation-bound statistics. The paper does not claim the transformer learned structure; it reports what the statistics do and do not transfer to.

Ongoing work and conflict declaration. The route does not end here. We are actively working on a direct formalization of the Riemann hypothesis through known mathematics, as the natural continuation of the Riemann direction that opens this route; and further directions of the framework are planned for publication. This paper claims only what is machine-checked within it; the concurrent and future work may overlap with parts of the route described here, and no priority is claimed beyond the timestamped evidence recorded in the accompanying manifest.

The proofs were developed in a human-AI collaboration. The direction, the framework claims, the observation judgments, and the corrections are human-authored; the Lean proof scripts were drafted by an AI assistant and checked by the machine. The collaboration is part of the

methodology, not of the proofs: every theorem stands on the machine check alone, and no claim depends on the drafting process. This mirrors the growing practice of AI-assisted formalization [9, 15, 18, 19].

Novelty discipline. The classical components are marked KNOWN: the interlock matrix is a 2–2 nonsingularity statement over the reals; the density of dyadic rationals in the interval is classical; the closure theorems are reformulations of standard density results in the framework’s vocabulary. The claims R136–R163 carry novelty_status annotations (KNOWN or NOVELTY_UNASSESSED) in the artifact’s claim ledger. The empirical observations are reported as observations, not as theorems.

9 Conclusion

The route from transformer empiricism to the Pat framework and the continuum is complete and recorded: the two-pole OOD dichotomy ($0/32 \leftrightarrow 32/32$, 45 runs reproduced) yields five structural laws (anchor binding, notation translatability, shift invariance, structure-bearing representation, definition-layer invisibility); the laws correspond observationally to the framework’s paired declaration, phase locking, and interlock; a completeness criterion (the decoupling operator D) organizes the correspondence; and the framework chain is machine-checked in Lean 4 / mathlib — twenty-two theorems, zero sorry, from the interlock matrix through the countable pat grid and its density to the continuum closure theorems.

In one sentence: the training layer says “representation-bound statistics,” the formalization layer says “paired-declaration structure,” and the two meet at the continuum closure: a representation is an anchor’s notation, and the continuum is the anchor’s closure.

References

- [1] Zhangir Azerbayev, Hailey Schoelkopf, Keiran Paster, Marco Dos Santos, Stephen McAleer, Albert Q. Jiang, Jia Deng, Stella Biderman, and Sean Welleck. 2023. Llemma: An Open Language Model For Mathematics. In *International Conference on Learning Representations (ICLR 2024)*.
- [2] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020. Language Models are Few-Shot Learners. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*.
- [3] The Lean Community. 2026. Lean 4 theorem prover. <https://lean-lang.org>
- [4] The mathlib Community. 2020. The Lean Mathematical Library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*.
- [5] Grégoire Delétang, Anian Ruoss, Jordi Grau-Moya, Tim Genewein, Li Kevin Wenliang, Elliot Catt, Chris Cundy, Marcus Hutter, Shane Legg, Joel Veness, and Pedro A. Ortega. 2022. Neural Networks and the Chomsky Hierarchy. In *International Conference on Learning Representations (ICLR 2023)*.
- [6] Shaoxiong Duan, Yining Shi, and Wei Xu. 2023. From Interpolation to Extrapolation: Complete Length Generalization for Arithmetic Transformers.
- [7] Jesse Michael Han, Jason Rute, Yuhuai Wu, Edward W. Ayers, and Stanislas Polu. 2021. Proof Artifact Co-training for Theorem Proving with Language Models. In *International Conference on Learning Representations (ICLR 2022)*.
- [8] Albert Q. Jiang, Sean Welleck, Jin Peng Zhou, Wenda Li, Jiacheng Liu, Mateja Jamnik, Timothée Lacroix, Yuhuai Wu, and Guillaume Lample. 2022. Draft, Sketch, and Prove: Guiding Formal Theorem Provers with Informal Proofs. In *International Conference on Learning Representations (ICLR 2023)*.
- [9] Haoan Lin, Zhiqing Sun, Sean Welleck, and Yiming Yang. 2024. Lean-STaR: Learning to Interleave Thinking and Proving.
- [10] Andreas Madsen and Alexander Rosenberg Johansen. 2019. Measuring Arithmetic Extrapolation Performance.
- [11] The mathlib Community. 2026. mathlib: The Lean mathematical library. Lean 4, mathlib v4.32.2.
- [12] Stanislas Polu, Jesse Michael Han, Kunhao Zheng, Mantas Baksys, Igor Babuschkin, and Ilya Sutskever. 2022. Formal Mathematics Statement Curriculum Learning. In *International Conference on Learning Representations (ICLR 2022)*.
- [13] Alethea Power et al. 2022. Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets. *arXiv preprint arXiv:2201.02177* (2022).
- [14] Ofir Press, Noah A. Smith, and Mike Lewis. 2021. Train Short, Test Long: Attention with Linear Biases Enables Input Length Extrapolation. In *International Conference on Learning Representations (ICLR 2022)*.
- [15] Peiyang Song, Kaiyu Yang, and Anima Anandkumar. 2024. Lean Copilot: Large Language Models as Copilots for Theorem Proving in Lean.
- [16] Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. 2021. RoFormer: Enhanced Transformer with Rotary Position Embedding. *Neurocomputing* (2021).
- [17] Ashish Vaswani et al. 2017. Attention is All You Need. *Advances in Neural Information Processing Systems* (2017).
- [18] Huajian Xin, Daya Guo, Zhihong Shao, Zhizhou Ren, Qihao Zhu, Bo Liu, Chong Ruan, Wenda Li, and Xiaodan Liang. 2024. DeepSeek-Prover: Advancing Theorem Proving in LLMs through Large-Scale Synthetic Data.
- [19] Kaiyu Yang, Aidan M. Swope, Alex Gu, Rahul Chalamala, Peiyang Song, Shixing Yu, Saad Godil, Ryan Prenger, and Anima Anandkumar. 2023. LeanDojo: Theorem Proving with Retrieval-Augmented Language Models. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*.
- [20] Kunhao Zheng, Jesse Michael Han, and Stanislas Polu. 2021. MiniF2F: a cross-system benchmark for formal Olympiad-level mathematics. In *International Conference on Learning Representations (ICLR 2022)*.